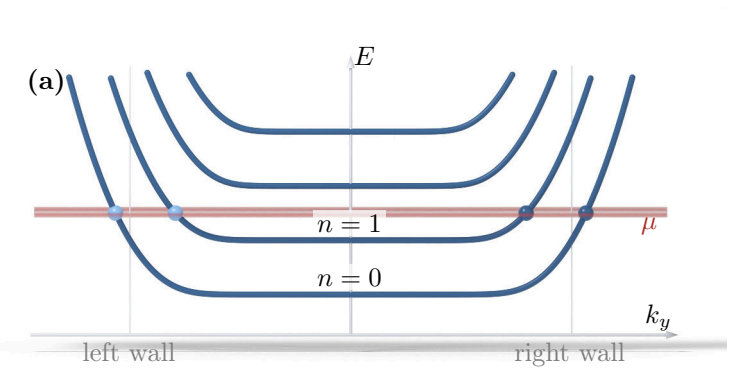
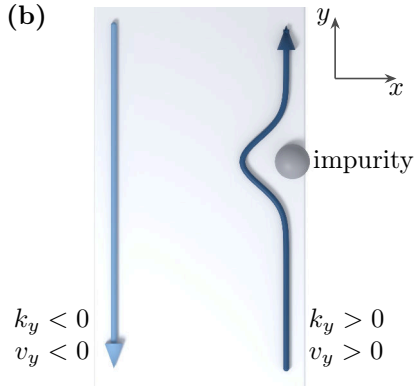

(a)
E
n = 1
μ
n = 0
k_y
left wall
right wall


(b)
y
x
impurity
k_y < 0
v_y < 0
k_y > 0
v_y > 0